Individual Project — Guidelines

Introduction to Machine Learning

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Description:

Important Information: Individual Work

Theme Submission:

Theme selection: Each student must select one of the topics provided.

Mandatory: The elements of the work-group (project 2) should select a theme from a different topic from each other.

Recommendation: Maximize the variability of selected themes within each class to provide a final portfolio that tackles different Data strategy areas. The submission is expected between February 14 and 20 (the theme must be validated with the professor).

Scope: A written theoretical/technical assignment in which students, individually, will explore a data-related topic. At the end of the semester, the various assignments will be compiled into a class portfolio on *"Data Compliance, Quality, and Exploration."*

The use of AI tools is allowed, however they have to be mentioned in the document, with preference given to those that enable reference tracking (e.g., scite.ai, consensus.app, elicit.com).

Evaluation: Evaluated based on document quality, technical/scientific accuracy, and critical analysis of the topic. There will be no presentation.

Document Requirements: Maximum of 4 pages, IEEE format.

Mandatory document sections:

- ✓ Critical Analysis - critical description of the selected topic, evidencing the student's reflection on the acquired knowledge
- ✓ Acknowledgement of used AI tools – including which tools had been used, the rationale behind the selection

Final Submission: 13/03/2026.

Goal: Develop written communication and critical thinking skills. Enhance the ability to interpret and assess the importance of data and its proper handling for the responsible development of ML tools.

Expected Contents in the Document:

- Abstract: Summary of the work.
- Introduction: Presentation of the topic, including references and a clear description of its relevance in the context of data processing and ML.
- Methodology: Critical description of existing methodologies within the topic.
- Discussion: Comparative and critical analysis of the studied methodologies.
- Final Section: The student should present their critical opinion on the chosen topic based on the studied methodologies.
- Acknowledgement of AI tools: Identify the AI tools used, describing the purpose of use, and the justification for the tool choice.

Topics in Study:

Topic 1 — Data Quality: Fundamentals and Diagnosis

Topic 2 — Data Exploration: Seeing, Understanding, and Questioning

Topic 3 — Human Data, Representativeness and Bias

Topic 4 — Decisions in Pre-processing and Their Implications

Topic 5 — Communication, Interpretation, and Narrative of Data

PL1

Topic 1 — Data Quality

1. Data Quality Issues in Small vs Large Datasets
2. Measurement Bias in Real-World Data
3. Temporal Drift and Data Degradation
4. Data Quality Metrics: What Can Be Measured?

Topic 2 — Data Exploration

5. Detecting Anomalies Through Visualization
6. Identifying Redundant Variables
7. Exploring High-Dimensional Data
8. Data Exploration for Hypothesis Generation

Topic 3 — Human Data & Bias

9. Demographic Imbalance and Its Consequences
10. Socioeconomic Bias in Public Datasets
11. Missing Populations in Datasets
12. Bias Amplification Through Models

Topic 4 — Pre-processing Decisions

13. Handling Imbalanced Data at the Data Level
14. Aggregation: Losing or Gaining Information?
15. Domain-Driven vs Data-Driven Preprocessing
16. Impact of Preprocessing on Downstream Models

Topic 5 — Communication

17. Telling the Story of a Dataset
18. Choosing Figures That Inform (Not Decorate)
19. How Data Exploration Communicates Assumptions and Decisions
20. Visual Consistency and Clarity

PL2

Topic 1 — Data Quality

1. What Does “Good Data” Mean in Machine Learning?
2. Missing Data: Patterns, Causes and Risks
3. Label Quality: The Hidden Weakness of Supervised Learning
4. Outliers: Errors, Rare Events or Valuable Signals?

Topic 2 — Data Exploration

5. Why Data Exploration Comes Before Modelling
6. Visual Exploration vs Statistical Summaries
7. Understanding Class Imbalance Through Data Exploration
8. Common Pitfalls in Exploratory Analysis

Topic 3 — Human Data & Bias

9. Representativeness in Machine Learning Datasets
10. Proxy Variables and Hidden Bias
11. Historical Bias Embedded in Data
12. Human Annotation and Subjectivity

Topic 4 — Pre-processing Decisions

13. Removing Missing Data vs Imputation
14. Encoding Categorical Variables: Impacts and Trade-offs
15. Overprocessing Data: When Cleaning Goes Too Far
16. Preprocessing Pipelines and Reproducibility

Topic 5 — Communication

17. Communicating Data Quality Issues Clearly
18. Explaining Limitations of a Dataset
19. Avoiding Misleading Interpretations
20. Ethical Communication of Data Findings

PL3

Topic 1 — Data Quality

1. Synthetic Data: Solution or New Problem?
2. The Cost of Ignoring Data Quality
3. Exploratory Data Analysis as a Quality Audit Tool
4. From Raw Data to Reliable Dataset

Topic 2 — Data Exploration

5. The Role of Scale and Normalization in EDA
6. Exploratory Analysis of Categorical Variables
7. When Exploration Conflicts with Intuition
8. From Exploration to Informed Decisions

Topic 3 — Human Data & Bias

9. Sensitive Attributes: To Include or Not?
10. The Role of Data Exploration in Bias Detection
11. Case Study Analysis of Biased Datasets
12. Responsible Use of Human-Centered Data

Topic 4 — Pre-processing Decisions

13. When Simplicity Improves Data Quality
14. Preprocessing as Part of Model Design
15. Comparing Alternative Preprocessing Strategies
16. The Role of Assumptions in Data Preparation

Topic 5 — Communication

17. Visualizing Uncertainty in Data
18. From Numbers to Meaning
19. Data Narratives and Decision-Making
20. Building Trust Through Honest Data Reporting

PL4

Topic 1 — Data Quality

1. Dimensions of Data Quality: Accuracy, Completeness, Consistency
2. Noise vs Signal: When Variability Is Not Information
3. Duplicate Data: Detection and Impact
4. When Cleaning Data Removes Important Information

Topic 2 — Data Exploration

5. Distribution Shapes and What They Reveal
6. Feature Distributions Across Classes
7. Multivariate Exploration: Beyond Pairwise Analysis
8. Reproducibility in Data Exploration

Topic 3 — Human Data & Bias

9. What Makes Human Data Different?
10. Sampling Bias in Real-World Data
11. Gender Bias in Data Collection
12. Data Quality vs Data Fairness

Topic 4 — Pre-processing Decisions

13. Data Cleaning as a Series of Decisions
14. Normalization Choices and Their Effects
15. Feature Selection as Information Filtering
16. Documenting Data Preparation Decisions

Topic 5 — Communication

21. Writing About Data Without Overclaiming
22. Data Exploration as Scientific Argument
23. Writing for a Non-Expert Audience
24. Common Communication Failures in ML Projects

PL5

Topic 1 — Data Quality

1. Human Errors in Data Collection
2. Inconsistent Encoding and Its Consequences
3. Data Quality in Public Open Datasets
4. Domain Knowledge as a Data Quality Tool

Topic 2 — Data Exploration

5. Correlation Is Not Explanation
6. Visual Bias: When Plots Mislead
7. Temporal Patterns in Sequential Data
8. Exploratory Analysis as a Communication Tool

Topic 3 — Human Data & Bias

9. Cultural Context and Data Interpretation
10. When Data Reflects Inequality
11. Ethical Risks of Poor Data Quality
12. Limitations of “Neutral” Data

Topic 4 — Pre-processing Decisions

13. Data Transformation and Interpretability
14. Discretization of Continuous Variables
15. Transparency in Data Preparation
16. Preprocessing Errors as a Source of Bias

Topic 5 — Communication

25. The Role of Context in Data Interpretation
26. Communicating Bias Without Sensationalism
27. Reproducibility as Part of Communication
28. Transparency vs Complexity in Data Reporting